

Workout Tracker Android App



Course

CSC-410 Mobile Application Development

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Instructor

Mam. Rabeeya Saleem

Department of Computer Science
University of Engineering and Technology, Lahore

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Contents

1	Executive Summary	2
2	Project Overview	3
2.1	Project Details	3
3	Key Features	4
3.1	User Authentication	4
3.2	Workout Management	4
3.3	Progress Analytics	4
3.4	Profile & Personalization	5
4	Technical Specifications	6
4.1	Software Stack	6
4.2	Project Structure	6
5	User Interface Design	7
6	Conclusion and Future Enhancements	8

1. Executive Summary

Workout Tracker is a comprehensive mobile application designed to help fitness enthusiasts effectively log their daily exercises, monitor their progress, and manage personal health data.

Built using modern Android development practices with Kotlin, the app emphasizes a clean, intuitive user experience and efficient data management. It provides secure multi-user support with complete data isolation, ensuring each user's workout history remains private.

The application serves as a personal fitness companion, enabling users to track workouts, analyze performance, and maintain their fitness journey with ease.

2. Project Overview

2.1. Project Details

- **Platform:** Android (Minimum SDK 24 - Android 7.0 Nougat)
- **Programming Language:** Kotlin
- **Architecture:** Singleton Repository Pattern
- **UI Framework:** XML Layouts with ViewBinding and Material Design 3

3. Key Features

3.1. User Authentication

- Secure multi-user support with account creation
- Complete data isolation — each user's workout data is private
- Persistent user session management throughout the app lifecycle

3.2. Workout Management

- Full CRUD (Create, Read, Update, Delete) operations for workout entries
- Detailed workout logging including:
 - Exercise Name
 - Number of Sets
 - Repetitions per set
 - Weight lifted (in kg or lbs)
 - Exercise Category (Strength, Cardio, HIIT, etc.)
- Smart empty state handling with helpful prompts and illustrations

3.3. Progress Analytics

- Real-time Summary Dashboard showing:
 - Total sets completed
 - Total repetitions
 - Total weight lifted
- Caloric Burn Estimation using the formula:

$$\text{Calories} \approx (\text{Sets} \times \text{Reps} \times \text{Weight}) \times 0.1$$

- Visual progress tracking for motivation

3.4. Profile & Personalization

- Comprehensive user profile management (Name, Age, Gender, Contact Information)
- Profile picture support — users can select image from device gallery
- Dynamic Navigation Drawer header that automatically syncs with latest profile updates

4. Technical Specifications

4.1. Software Stack

- **IDE:** Android Studio
- **Build System:** Gradle with Kotlin DSL
- **UI Components:** AppCompatActivity + Fragments + Material Design 3
- **Data Storage:** In-memory List-based storage (optimized for fast performance during development and testing)

4.2. Project Structure

The project follows a clean and modular package structure:

- `com.example.workouttracker.data` — Repository layer (`AppRepository`, `WorkoutRepository`)
- `com.example.workouttracker.model` — Data models (`User`, `Workout`)
- `com.example.workouttracker.adapter` — Custom RecyclerView adapters
- `com.example.workouttracker.ui` — All UI components (Activities and Fragments for Authentication, Dashboard, Workout Entry, and Summary)

5. User Interface Design

The application strictly follows **Material Design 3** guidelines to deliver a modern and intuitive user experience.

Key UI elements include:

- **Navigation Drawer** — for seamless access to all major sections
- **CardView** components — for clean and organized display of workout entries
- **Floating Action Button (FAB)** — for quick access to primary actions (e.g., adding a new workout)
- **ConstraintLayout** — ensuring responsive and adaptive layouts across different screen sizes and orientations
- Consistent use of colors, typography, and spacing as per Material Design principles

6. Conclusion and Future Enhancements

Workout Tracker successfully demonstrates the implementation of core Android development concepts including the repository pattern, secure user management, RecyclerView optimization, and real-time analytics calculation.

Possible Future Enhancements:

- Integration with Room Database for persistent local storage
- Cloud synchronization using Firebase Firestore
- Workout templates and exercise library
- Integration with wearable devices (Google Fit / Health Connect)
- Advanced charts and graphs using MPAndroidChart
- Dark mode support and theme customization
- Export workout data as PDF or CSV

This project serves as a solid foundation for a production-ready fitness tracking application.